Peer Review — Off-centre CoE cosmology (main_rev5)

ARS academic-paper-reviewer workflow; NOT_CALIBRATED; single-executor sequential passes. Contents: editorial decision & top-10 issues; math-check table; five reports; Phase-0 field analysis.

Editorial Decision

Manuscript: Off-centre CoE cosmology: mathematical workflow (main_rev5.tex)

Decision: MAJOR REVISION (as a journal article it would be Reject-and-resubmit; the three CRITICAL items must be resolved before re-review)

Calibration: NOT_CALIBRATED. Provenance: all five passes were written sequentially by one executor following the ARS academic-paper-reviewer workflow; they are not independent human reviews.

Blocking Issues (3)

1. CMB isotropy for a 58-Gpc off-centre observer not demonstrated (DA-2, R2-W1). [CONSENSUS-3: DA, R2, EIC]

2. Cluster χ^2 on correlated bins; null not rejected; generic harmonics fit better (DA-1, R1-W1/W2, R3-W1). [CONSENSUS-4]

3. Inconsistent cosmological background ($H_0 \approx 47$ vs 70 vs ~145; ΛCDM dark-matter inputs) (DA-3, R2-W2/W3). [CONSENSUS-3]

Reviewer Summary

Reviewer	Recommendation	Confidence
EIC (journal fit)	Major Revision	4
R1 Methodology	Major Revision	4
R2 Domain	Major Revision (Reject-leaning for CMB)	4
R3 Perspective	Major Revision	3
Devil's Advocate	Reject in current form	4

[SPLIT] Severity of the CMB acoustic fit: EIC/R3 treat it as a legitimate "can it mimic" exercise if reframed; R2/DA treat it as evidence against the model. Editor: reframe, and state $\Delta\chi^2 \approx 1{,}229$ as disfavouring.

Top 10 issues (ranked)

#	Severity	Location	Problem	Concrete fix
1	Critical	§4.3, §4.5	Residual geometric CMB multipole after 95.9 % cancellation ≈ 4e-4–2e-3 ≫ 1e-5; off-centre LTB bounds (observer within ~tens of Mpc of centre) not addressed	Compute ℓ=1–3 ΔT/T of the full final sky; confront Alnes & Amarzguioui 2006, Foreman et al. 2010
2	Critical	§3.2, §4.1, Abstract	71 overlapping 90° cones at 5° steps (~4 independent) fit as independent; χ^2/dof ≈ 0.11–0.17; null P=0.993	Per-cluster likelihood on the 313-cluster catalogue or bootstrap 71×71 covariance
3	Critical	Eq. 2, Eq. 5, §4.4, Table 1	t_0=13.8 Gyr w/o Λ ⇒ H≈47; lock uses H_0=70; R=4 Gpc ⇒ H_0≈145; web uses Ω_m=0.31 + CDM transfer	One baryon+radiation background; derive a_eq, R, growth, σ_8 from it
4	Major	§4.1, Fig. 7	3-harmonic Fourier (χ^2=3.17) beats all CoE laws (7.67–11.6)	Model comparison (AIC/BIC/CV) vs n=1–3 templates and a bulk-flow model
5	Major	§4.7.2, Abstract	11/12 sign score: post-hoc orientation/smoothing/assignment, correlated antipodal spots, no null (naïve binomial 0.0032 invalid)	Rotation/assignment Monte Carlo; one pre-fixed orientation
6	Major	Eq. 19, Table 2	Acoustic peaks inserted via cos(k r_s), r_s free; χ^2 3,787 vs 2,558 (χ^2/dof 1.52 vs 1.03)	Reframe as mimicry test; blind high-ℓ / TE/EE prediction

#	Severity	Location	Problem	Concrete fix
7	Major	§4.5; run9	Tidal shear/lensing omitted from TT stage; shear step not reported and not reproduced (48.9 vs 149.9)	Add lensing; pin seeds/code; report and reproduce shear step
8	Major	Eq. 4	CatWISE dipole (kinematic+clustering of z≈1–2 sources) equated to density contrast at D±14.3 Gpc	Integrate model density+velocity over CatWISE dN/dz
9	Major	Fig. 18 caption, Table 2	3,787 is the ensemble mean; seed-42 realisation gives 4,601 (2–30: 148)	Relabel; compare data with realisation distribution
10	Major	§1, refs; Abstract	Motivating SN acceleration never computed; LTB/kSZ/μ-distortion literature absent	Compute $D_L(z,\hat{n})$ vs Pantheon+; add related work

Further minor: Eq. 19 example 2.2e-9 → 2.06e-9; Planck peak lists disagree (224/535/821/1122/1412 vs 228/544/823/1121/1417); D=42 vs 58.2 in web; R=8 labelled PASS despite lowl_over 2.99>2.5 (and R=14 FAIL unstated); s_w on bound; ℓ notation clash; χ^2 error convention; unbinned vs binned data; δ rounding (L 920.3 vs 922.65); incomplete references; f, p(s), K_G unspecified.

DA-CRITICAL disposition

DA-1, DA-2, DA-3 accepted as CRITICAL (data–conclusion mismatch; foundation collapse; logic-chain break). DA-4–6 accepted as Major.

Rationale

The arithmetic is sound (≈35 of 45 checks verify; see Appendix M) and the manuscript is unusually candid. But the inferential claims (cluster improvement, sky-anomaly agreement) are not supported by valid statistics, the model is not internally consistent, and the most stringent test (CMB isotropy) is unaddressed. These are fixable in principle but require new analysis, not edits.

Roadmap

1. Fix the background (Issue 3). 2. Full-sky low-ℓ CMB (Issue 1). 3. Cluster likelihood + model comparison (2, 4). 4. Anomaly null (5). 5. Reproduce shear, add lensing (7). 6. Reframe TT fit and relabel (6, 9). 7. Minor list.

Appendix M — Mathematics Check

Method: each equation re-derived by hand; numbers recomputed in mathcheck.py (this folder) and spot-checked against box files. "Verified" = re-derived and/or reproduced to the quoted precision. "Error" = wrong as written. "Unverified" = not checkable from the paper or box in this review. No code was re-run except the small checks in mathcheck.py; the run9 rerun result is the earlier rerun's log (/workspace/coepdf/run_out/run9/run9.log), not re-executed here.

#	Item (location)	Status	Note
M1	Volumes, §2.2 (V_R=268.1, V_{D+R}=1.008e6, 0.0266 %, 1.618e6, 0.0166 %)	Verified	268.08; 1.00800e6; 0.02660 %; 1.61815e6; 0.01657 %. 72.83 ≠ 1.25×58.2=72.75 (paper already notes).
M2	Eq. 1 ρ_r/ρ_m0 = a_eq/a⁴; example a=1e-4 → 2.94e12	Verified	Standard.
M3	Eq. 2 closed form of ∫a′da′/√(a_eq+a′)	Verified	Closed form = quadrature to 1e-11 (a=0.5). Limits a^{1/2}, a^{2/3} correct.
M4	t(a_eq)=40.8 kyr, t(1/1090)=297.5 kyr with t(1)=13.8 Gyr	Verified (arithmetic)	40.79 kyr, 297.47 kyr.
M5	Implied H_0 of Eq. 2 normalisation	**Error (inconsistency)**	Forcing t_0=13.8 Gyr with no Λ gives H* ≈ 47.2 km/s/Mpc. The CatWISE lock uses H_0=70, Ω_m=0.3 (EdS age at H_0=70 is 9.31 Gyr). The thermal history and the density normalisation use incompatible H_0. a_eq=1/3400 is itself a ΛCDM value (depends on Ω_m h² including CDM).
M6	Recombination radius R = 4 (8, 14) Gpc	**Error / unsupported**	R is a free input, not derived. Matter-only comoving distance to z=1090: 8.31 Gpc (H_0=70) or 12.37 Gpc (H_0=47.2). χ*=4 Gpc would need H_0≈145. The final CMB fit (R≈4, R_c=3.57 Gpc) is not tied to the model's own kinematics.
M7	Volume growth n/t; 2/13.8 = 0.145 Gyr⁻¹	Verified	
M8	CoE kinematics: r(s,n̂)=√(D²+s²+2Ds cosψ), first-order dipole in cosψ	Verified	Correct; note expansion velocity at D=58.2 Gpc with H_0=70 is ≈13.6c — Newtonian kinematics invalid there (paper concedes "superluminal").
M9	Eq. 3 A/A_all functional	Unverified	f and p(s) never specified (paper self-flags). The forward map cannot be reproduced from the text.
M10	Eq. 4 δ=0.031 = 2×CatWISE dipole	Arithmetic verified; **physics error**	2×0.01554 = 0.03108 (rounded to 0.031: L becomes 920.3 vs 922.65 Gpc, negligible). But a number-count dipole is mostly kinematic (∝[2+x(1+α)]β) plus a clustering term integrated over the quasar redshift range (z≈1–2, i.e. few Gpc), not a density difference at D±14.3 Gpc. Equating it to [ρ(D−d)−ρ(D+d)]/ρ(D) is an unjustified identification.
M11	Eq. 5 L=922.65 Gpc; ρ_0=0.3195; ρ′/ρ=−0.001084; d=3 → 193.56	Verified	922.650; 0.31953; −0.0010838; 193.556. Matches catwise-density-levelset-fit.json.
M12	Law table (power, cored, Gauss, soft, stretched, η-edge): δ=0.031 and ρ_0/ρ_c	Verified	All reproduce δ=0.0310 at the quoted L (to the 4-5 s.f. given); ρ_0 values 0.3201, 0.3098, 0.3096, 0.3784, 0.3847, 1.0319 reproduced.
M13	"all laws ρ′/ρ ≈ −0.00106 to −0.00109 Gpc⁻¹"	Verified	At D: −0.00108 (exp, power), −0.00109 (cored, Gauss), −0.00107 (soft, stretched), −0.00106 (η-edge).
M14	Null χ² = 45.14 over 71 bins	Verified	45.137 using mean of asymmetric errors (fig6_digitized.csv). With low/high/side errors: 60.5 / 40.7 / 43.4 — the choice matters and is not stated. Note Eq. 9's side rule is not what produced 45.14.
M15	χ²/dof and significance of 45.14 → 7.67–11.6	**Error in interpretation**	Treating bins as independent: P(χ²≥45.14 | 71) = 0.993, i.e. the null itself is "too good". Best fits give χ²/dof ≈ 0.11 (7.67/68), 0.16 (10.79/67), 0.17 (11.6/69); P(χ²≤7.67|69) ≈ 2e-21. Bins are 90° sliding cones at 5° steps — each cluster enters ~18 bins; effective independent bins ~4. χ² values cannot be read as fit quality or significance. No significance is quoted in the paper, but "χ² falls from 45 to 10–12" is presented as the strongest evidence.
M16	A_F baseline (AF_smooth)	Verified (it is the axisymmetric CoE model, not a Migkas fit)	3-harmonic Fourier series reproduces AF_smooth to 1.5e-5 rms; a 3-harmonic fit to Migkas A differs from AF by 0.055 rms. **But**: data vs AF χ² = 33.2 (71 bins, mean errors); a free 3-harmonic (7-parameter) Fourier fit reaches χ² = 3.17. Generic low-order harmonics beat every CoE law (7.67–11.6).
M17	m(r) modulation (Fig. d5): slope −9.8e-4 at r=42, m(56.3)=0.987, R_b=2.011	Verified	c_0/42 = −9.76e-4; 0.9874; 2.0107. **Inconsistency**: web uses D=42 Gpc, cluster fit D=58.2 (refits 35–60 Gpc).
M18	Eq. 6–7 harmonics, D_ℓ; example 1.30e4 μK²; θ≈180°/ℓ → 0.82° at ℓ=220	Verified	13005 μK².
M19	Eq. 8 cosmic variance values	Verified	0.632, 0.309, 0.181, 0.0673, 0.0316.
M20	Eq. 9 asymmetric-error χ²	Verified as stated (convention ok)	Side-switching makes χ² non-smooth in parameters; fine as comparison only (paper says so).

#	Item (location)	Status	Note
M21	Eq. 10 Pearson r^2	Verified	Correct; r^2 comparison 0.976 vs 0.983 is uninformative (paper acknowledges).
M22	Eq. 11 Press–Schechter, bias $b=1+(\nu^2-1)/\delta_c$	Verified (standard)	But uses ΛCDM $\sigma_8=0.8$, $\Omega_m=0.31$ with Eisenstein–Hu CDM transfer: dark matter enters the "no dark matter" model.
M23	Void deficit $R_v=30$ Mpc → 4.0e15 $M_\odot$	Verified	4.05e15.
M24	Eq. 13 tidal signal s_R and K_G scaling	Unverified	$K_G=7.7e51/0.0039$ physical origin unknown (self-flagged). s_R is a radial *acceleration* difference, not a temperature; the map is rescaled by ε fitted per R, so its amplitude is free.
M25	Eq. 14 ladder values for universe #3 (ε, χ^2)	Verified vs guesses.json	29.0 %/441.6, 25.0/411.5, 25.0/177.2, 21.6/69.6, 52.5/85.0, 45.3/47.2. **Confirmed inconsistency**: R=8 lowl_over=2.994 > 2.5 yet "PASS"; also universe #0 R=8 (2.91, PASS). R=14 for #3 is FAIL (slope 0.060>0.05) — paper does not state this.
M26	run9 shear step (272,175 bodies; pattern/noise 149.9)	**Not reproduced**	Earlier rerun log: 272,638 bodies, 48.9 vs repo 149.9. Cause unknown. Shear step is not reported in the paper at all, yet is the first fit step of the method.
M27	Eq. 15 Poisson; $4\pi G\bar{\rho}$ = 2.278e9; Φ/c^2 = 2.5e-6	Verified	2.2777e9; 2.534e-6.
M28	SW $\Phi/3c^2$, ISW $(2/c^2)\int\Phi dt$	Verified (standard linear GR, but inserted into a Newtonian superluminal flow)	
M29	$\langle(\hat{k}\cdot\hat{u})^2\rangle=1/3$	Verified	
M30	Eq. 17 exact $C_\ell = (2/\pi)\int k^2 dk\, P_\Phi \lvert\Delta_\ell\rvert^2$	Verified	Prefactor re-derived from the stated Fourier convention.
M31	Eq. 18 Limber	Verified (form)	Standard; W′ term dropped in Limber part (not stated).
M32	R_c/w = 10.6; source 1.28w above centre	Verified	10.645; 1.277.
M33	Eq. 19 example $P(k=100)$ = 2.2e-9 Gpc^3	**Error (minor)**	With the damping factor stated ($e^{-0.089}$) it is 2.06e-9; 2.2(5)e-9 is the value *without* damping.
M34	Eq. 20 $\Delta\ell$ = 302.6, 282; χ_eff 3.8; 57.1/114.2/199.9 Mpc	Verified	302.57, 282.27, 3.796, 57.1/114.2/199.9.
M35	Residual mapping r_2=41.7 Mpc, k_B=251 Gpc^{-1}	Verified	41.75, 251.3.
M36	Fit table (3787/0.976/137; 4131/0.966/49.1) and parameters	Verified vs acoustic_half_fixed.json	All match. Note: 2499 unbinned points used (npts), not the binned $\ell\geq30$ file the Data section implies.
M37	χ^2 3787 vs ΛCDM 2558	Verified; interpretation added	$\Delta\chi^2$=1229 with 12 vs ~6 parameters; χ^2/dof 1.52 vs 1.03 — CoE decisively disfavoured (paper says "short of ΛCDM", understates). The 3787 is the **ensemble mean** spectrum; the seed-42 one-sky realisation gives 4601 (2–2500) and 148 (2–30), but Fig. 18 caption calls the red curve a one-sky realisation with χ^2=3787.
M38	Planck peak positions	**Error (internal inconsistency)**	§Acoustic fit: Planck 224, 535, 821, 1122, 1412; Table 2: 228, 544, 823, 1121, 1417. Same data, two lists.
M39	"half" variant parameters	Verified; flag	s_w=0.0500 sits on its lower bound (boundary solution) — not disclosed.
M40	$\Delta\chi^2$=2.3, 6.2, 11.8 for 2 parameters	Verified	2.296, 6.180, 11.829.
M41	Harmonic decomposition A: 25.54→15.30→12.06→11.46; B 13.68→8.67→10.04→10.24; phases C_2=109.0° ↔ peaks 54.5/234.5, C_3=248.2° ↔ 82.7°	Phases verified (109/2=54.5, 248.2/3=82.7) ; χ^2 values unverified	Scripts dumpM.py/recon_n3.py/cum2.py not re-run. n=1 peak at "ℓ=102.8°" while C_1 phase 102.8° — consistent. Notation clash: ℓ used for Galactic longitude and multipole.
M42	11/12 sign agreement	Arithmetic: binomial $P(\geq11/12 \mid p=0.5) = 0.0032$ — **but not a valid significance**	Spots are not independent (antipodal pairs: dipole hot/cold, bulk flow toward/away), orientation (15° lift, 45° rotation) and ±25° smoothing chosen after the fact, A/B assignment is the author's, n=1–3 removal chosen. No null simulation.
M43	CMB isotropy: "95.9 % cancellation" of a several-percent geometric multipole	**Error in sufficiency**	Remaining residual = 4.1 % of 1–5 % ≈ 4e-4–2e-3, i.e. 40–200× the observed ~1e-5 low-ℓ anisotropy. Literature bounds (Alnes & Amarzguioui 2006, PRD 74 103520; Foreman, Moss, Zibin & Scott 2010, PRD 82 103532) put an observer in a Gpc-scale LTB structure within ~tens of Mpc of the centre ($\lesssim$ ~1 % of the void radius) from the CMB dipole/quadrupole — here the observer is 58 Gpc off-centre. [exact literature numbers quoted from memory — verify before citing]

#	Item (location)	Status	Note
M44	Power-spectrum stage omits tidal shear	Verified (code)	fwd.py/acoustic_model.py: SW+ISW+moving lens only; moving lens ≥99.99 %. Lensing/shear not in the TT fit; the shear (run9) step is separate and not reproduced (M26).
M45	H/H_ref = 1/√(A/A_all)	Verified	Follows from $A \propto H_0^{-2}$ at fixed flux.

Peer Review Report — Journal-Fit Reviewer (EIC)

Manuscript Information

Off-centre CoE cosmology: the mathematical workflow from geometry to the Planck TT spectrum (rev. 5). Reviewed as a submission to a PRD/JCAP-type journal.

Reviewer Information

• **Role:** Journal-Fit Reviewer (card #1). **Identity:** editor handling non-standard cosmologies. **Focus:** originality, significance, framing, fit.

Overall Assessment

• **Recommendation:** Major Revision (not publishable as a research article in current form; acceptable as an arXiv working note after reframing).

• **Confidence:** 4/5. **Calibration:** NOT_CALIBRATED.

• **Summary:** An honest, unusually self-critical working document that walks from an off-centre expanding-matter geometry to fits of the Migkas cluster anisotropy and the Planck TT spectrum. Its candour (self-flags, "the question is posed, not answered", CMB fit worse than ΛCDM) is a strength. But it is structured as lab notes, not a paper: it has no stated falsifiable claim, the central motivation (apparent acceleration from SNe) is never computed, and it does not engage the large literature on off-centre LTB observers that already constrains exactly this set-up.

Strengths

• **S1 Transparency.** Every number is tied to a file; unverified items are flagged in-line (\flag).

• **S2 Honest CMB result.** States χ^2=3,787 vs 2,558 for ΛCDM and that the half-scale mode has no physical origin (§Discussion).

• **S3 Clear derivations** of standard tools (Eqs. 6–10, 15–20) — useful pedagogically.

Weaknesses

• **W1 [MAJOR] No engagement with prior off-centre/LTB literature.** Location: §1, References. Problem: Alnes & Amarzguioui (2006), Foreman et al. (2010), Clifton–Ferreira–Land (2008), Zibin–Moss–Scott kSZ tests (2008/2011), Bull–Clifton–Ferreira (2012), Caldwell & Stebbins (2008) all test off-centre/void models against the CMB. Fix: add a related-work section positioning the CoE model against these constraints.

• **W2 [MAJOR] Motivating claim untested.** Location: Abstract, §2.1, §5 next steps (5). Problem: the paper is motivated by apparent acceleration from SNe distances, yet D_L(z) is never computed. Fix: compute D_L(z, n̂) along off-centre lines of sight and compare with Pantheon+/Union3, or drop the motivation from the abstract.

• **W3 [MAJOR] Framing of the cluster result.** Location: Abstract, §4.1, Conclusions bullet 1. Problem: "χ^2 falls from 45.14 to about 10–12" is presented as the strongest point, but the bins are overlapping sliding cones; no significance follows. Fix: present with a proper likelihood (see methodology).

• **W4 [MINOR] Reference list incomplete** (no years/journal/volume for Ellis & Stoeger, Migkas, Planck, Eisenstein & Hu, Press–Schechter). Fix: full citations (Migkas et al. 2020 A&A 636 A15; Secrest et al. 2021 ApJL 908 L51; Planck 2018 VI; etc.).

• **W5 [MINOR] Title/Author.** "Working notes for C. A." and figures "restored from the original preprint" signal a draft; a journal needs a self-contained figure set with reproducible code.

Detailed Comments

• **Title & Abstract:** the abstract reports 7 data numbers but no conclusion about viability; state plainly that the CMB stage disfavours the model at $\Delta\chi^2 \approx 1{,}200$.

• **Introduction:** good statement of question; missing related work.

• **Methods/Results:** too long on textbook derivations (D_ℓ, cosmic variance, Limber) relative to model-specific content; move to an appendix.

• **Discussion:** the "Tensions" paragraph is the paper's real conclusion and should lead.

Questions for Authors

1. What observation would falsify the CoE model in your view?

2. How does the model evade the Alnes/Foreman off-centre bounds?

3. Why are two different sky orientations (+80.8° roll; 45° rotation) both used?

Minor Issues

• ℓ used both for Galactic longitude and multipole (§4.7). Two Planck peak lists disagree (§4.5 vs Table 2).

Criterion-Bound Judgements

Criterion	Judgement
Originality	Moderate (off-centre matter geometry + cluster anisotropy combination is uncommon; CoE/LTB idea is not new)
Significance	Low as presented (no viability claim survives the CMB stage)
Journal fit	Poor as a research article; fine as arXiv notes
Clarity	Good locally, poor globally (no throughline)

Peer Review Report — Reviewer 1 (Methodology)

Reviewer Information

• **Role:** Methodology (card #2). **Identity:** cosmological statistician. **Focus:** likelihoods, correlated data, dof, model comparison, reproducibility.

Overall Assessment

• **Recommendation:** Major Revision. **Confidence:** 4/5. **Calibration:** NOT_CALIBRATED.

• **Summary:** The arithmetic is largely right (see Appendix M), but the inferential layer is not. The cluster χ^2 is computed on 71 heavily overlapping sliding-cone bins as if independent; the 11/12 anomaly score has no null distribution; the CMB stage compares an ensemble-mean spectrum to one sky while labelling it a realisation; and the method's first step (run9 shear) does not reproduce.

Strengths

• **S1** Parameters and χ^2 values in Table 2 reproduce exactly from acoustic_half_fixed.json (M36).

• **S2** Cosmic-variance treatment (Eq. 8, χ^2_CV) is correct and its limits stated.

• **S3** Limber validity is quantified against the exact projection (acoustic_limber_check*.json).

Weaknesses

• **W1 [CRITICAL] Correlated Migkas bins treated as independent.** Location: §3.2, §4.1 "Fit", Abstract. Evidence: Migkas Fig. 6 cones have $\Delta l = 90°$ and are sampled every 5°, so each cluster contributes to ~18 adjacent bins; effective independent bins ≈ 360/90 = 4. Under the independence assumption the null $\chi^2 = 45.14/71$ has P=0.993 (the null is "too good"), and fits give $\chi^2/\mathrm{dof} \approx 0.11$–$0.17$ (M15). Consequence: χ^2 differences 45→7.67 have no probabilistic meaning; with ~4 effective points, a 2–4 parameter model fitting them well is expected. Fix: refit the 313-cluster catalogue (migkas2020_tableC1.csv is on the box) directly: per-cluster L_X–T likelihood with a direction-dependent $A(\hat{n})$, or build the full 71×71 bin covariance by bootstrap over clusters and use $\chi^2 = r^T C^{-1} r$.

• **W2 [MAJOR] No comparison against generic low-order models.** Location: §4.1, Fig. 7. Evidence: a free 3-harmonic Fourier series (7 params) gives $\chi^2 = 3.17$ on the same 71 bins; the paper's own best "variant" (7.67) is said to equal a "reference sinusoid". Fix: report AIC/BIC or cross-validated likelihood of every CoE law against n=1,2,3 Fourier baselines; the CoE geometry must beat equally flexible phenomenology to count as evidence.

• **W3 [MAJOR] 11/12 sky-anomaly score lacks a null and has researcher degrees of freedom.** Location: §4.7.2, Fig. 22, Abstract. Evidence: orientation (15° lift, 45° rotation), smoothing (±25°), harmonic order (n=1–3), exclusion regions and A/B assignment are all choices; antipodal spots (dipole poles, bulk-flow toward/away) are anti-correlated by construction; one spot sits on zero (+0.02 %). Naïve binomial P=0.0032 is not valid. Fix: Monte-Carlo the score over random rotations of the residual map (and random spot assignments), report the fraction scoring ≥11/12; pre-register the orientation.

• **W4 [MAJOR] Ensemble mean vs realisation.** Location: Fig. 18 caption, Table 2. Evidence: $\chi^2 = 3{,}787$ is the ensemble-mean spectrum; the seed-42 one-sky realisation gives 4,601 (2–2500) and 148 (2–30) (M37). Fix: label correctly; compare the observed sky to the distribution of realisations.

• **W5 [MAJOR] First fit step not reproduced and not reported.** Location: absent (method §4.4/4.5 never presents the total-shear run9 step). Evidence: rerun gives 272,638 bodies and pattern/noise 48.9 vs recorded 272,175 / 149.9 (M26). Fix: pin code versions/seeds, archive inputs, report the shear step and its sensitivity to seed and thinning.

• **W6 [MAJOR] Parameter counting and boundary solutions.** 12 CMB parameters + D, l_0, shape, tilt, gradient, roll, rotation; "half" fit has s_w on its bound (0.050) (M39). The "fixed" sky placement (tilt/gradient/roll) is described as adding "no morphological degree of freedom", yet the 80.8° roll and 45° rotation are chosen values. Fix: count them as parameters; report bounds hit.

• **W7 [MINOR] Error convention.** 45.14 uses the mean of asymmetric errors; Eq. 9's side rule gives 43.4 (M14). State it.

• **W8 [MINOR] Data file.** §3.3 says binned file for $\ell \geq 30$, fit uses 2,499 unbinned points (M36).

• **W9 [MINOR] Many results "taken from earlier figure set; script not re-run"** (harmonic amplitudes, $\ell = 50$ step, $\Delta\ell \approx 1384$). Re-run or remove.

Questions for Authors

1. What is the bin-to-bin correlation matrix of the digitised Fig. 6?

2. What fraction of random map rotations give ≥11/12?

3. Which code revision produced the repo's 149.9?

Criterion-Bound Judgements

Criterion	Judgement
Design validity	Weak (inference on correlated bins)
Statistical reporting	Inadequate (no significance, no covariance, no model-selection penalty)
Reproducibility	Partial (CMB numbers reproduce from JSON; shear step does not; several scripts missing)

Peer Review Report — Reviewer 2 (Domain)

Reviewer Information

• **Role:** Domain (card #3). **Identity:** CMB / inhomogeneous-cosmology theorist. **Focus:** physical consistency, CMB isotropy, literature.

Overall Assessment

• **Recommendation:** Major Revision (bordering Reject for the CMB section as a physical model). **Confidence:** 4/5. **Calibration:** NOT_CALIBRATED.

• **Summary:** The model mixes three incompatible frameworks: Newtonian kinematics at distances where recession is ≫c, an FRW thermal history normalised to an age that implies $H_0 \approx 47$, and ΛCDM inputs ($\Omega_m = 0.31$ including dark matter, σ_8, Eisenstein–Hu CDM transfer, $a_{eq} = 1/3400$) in a model stated to have no dark matter. The CMB spectrum is then reproduced by a phenomenological P_δ with an acoustic-like oscillation and a second damped mode inserted by hand, so the acoustic peaks are put in, not predicted.

Strengths

• **S1** Correct statement that geometry alone gives a percent-level CMB multipole (§4.3) — consistent with LTB literature.

• **S2** Correct linear SW/ISW/Limber machinery (M28–M31).

• **S3** The paper recognises that the moving-lens regime is superluminal and treats it as an assumption.

Weaknesses

• **W1 [CRITICAL] CMB isotropy for an observer 58 Gpc off-centre.** Location: §4.3. Evidence: even after the stated 95.9 % cancellation, a 1–5 % geometric multipole leaves 4e-4–2e-3 (M43), 40–200× the observed ~1e-5. Off-centre LTB studies (Alnes & Amarzguioui 2006; Foreman et al. 2010; Quartin & Amendola 2010) restrict the observer to within tens of Mpc of the centre of a Gpc structure using dipole/quadrupole/octopole. Fix: compute the full dipole–quadrupole–octopole of the final model's CMB sky (geometry + Doppler + gravitational redshift + web), not just the stochastic part's power spectrum, and confront the off-centre bounds explicitly.

• **W2 [CRITICAL] Internal inconsistency of cosmological inputs.** Location: §2.2 Eq. 2, §4.1 CatWISE lock, §4.4 ingredients. Evidence: $t_0 = 13.8$ Gyr without Λ requires $H_0 \approx 47$ km/s/Mpc; the lock uses $H_0 = 70$, $\Omega_m = 0.3$; the web uses $\Omega_m = 0.31$, $\Omega_b = 0.049$ (i.e. 84 % non-baryonic matter) and a CDM transfer function (M5, M22). Fix: pick one background with only baryons + radiation (Ω_b-only), derive a_{eq}, $t(a)$, growth, transfer function and σ_8 from it, and redo the web.

• **W3 [MAJOR] Recombination radius not derived.** Location: Table 1, §2.1, §4.5. Evidence: R=4 Gpc requires $H_0 \approx 145$ in a matter-only model; the model's own value would be 8–12 Gpc (M6). Fix: derive R from the CoE kinematics/D(t) and keep it fixed in the TT fit.

• **W4 [MAJOR] Acoustic peaks inserted by parametrisation.** Location: Eq. 19. Evidence: A_w cos(k r_s+φ) and B e^{-k/k_B}cos(k r_2+φ_2) supply the peak structure; r_s=79.5 Mpc is free. In ΛCDM the peaks, their odd/even heights (baryon loading) and Silk damping are predicted from $\Omega_b h^2$, $\Omega_c h^2$. Fix: state clearly that the TT fit tests whether a parametrised source can mimic the spectrum, not whether the CoE model predicts it; or derive P_δ from baryon-photon physics in the CoE background.

• **W5 [MAJOR] Tidal shear/lensing absent from TT stage.** Location: §4.5. Evidence: fwd.py uses SW+ISW+moving lens only (M44); the shear step (run9) is separate and unreported. Planck measures CMB lensing ($C_\ell^{\varphi\varphi}$) and peak smoothing consistent with ΛCDM at the few-% level. Fix: add lensing of the primary spectrum and predict $C_\ell^{\varphi\varphi}$; compare with Planck lensing.

• **W6 [MAJOR] Missing observables.** Polarisation (TE/EE), kSZ (Zibin & Moss 2011 constraints on large-scale radial flows, especially with a ≥4c moving-lens flow), μ/y spectral distortions (Caldwell & Stebbins 2008), BAO, SN $D_L(z)$. Fix: at least discuss kSZ and spectral distortions, which are the decisive LTB tests.

• **W7 [MAJOR] CatWISE lock physics.** Location: Eq. 4. Evidence: the number-count dipole is a kinematic + clustering signal of sources at z≈1–2, integrated along the line of sight, not ρ(D−d)−ρ(D+d) at 14 Gpc (M10). Fix: model the predicted CatWISE dipole by integrating the CoE density and velocity over the quasar dN/dz with α, x from Secrest et al.

• **W8 [MINOR]** Follow-up dipole work (Secrest et al. 2022, NVSS+CatWISE) and Migkas et al. 2021 (bulk-flow interpretation of the same anisotropy) are relevant and absent. [citations from memory — verify details]

Questions for Authors

1. What is $\Delta T/T$ at $\ell=1,2,3$ for the full final model sky?

2. What fixes $r_s=79.5$ Mpc physically?

3. How is a superluminal Newtonian moving-lens kernel justified versus a relativistic LTB ray-trace?

Criterion-Bound Judgements

Criterion	Judgement
Theoretical consistency	Fails (mixed frameworks)
Literature coverage	Inadequate
Contribution	Exploratory; no predictive CMB result

Peer Review Report — Reviewer 3 (Perspective)

Reviewer Information

• **Role:** Perspective (card #4). **Identity:** inverse-problems / applied mathematician. **Focus:** identifiability, model flexibility, what the fits can and cannot show.

Overall Assessment

• **Recommendation:** Major Revision. **Confidence:** 3/5. **Calibration:** NOT_CALIBRATED.

• **Summary:** Viewed as an inverse problem, each stage of the pipeline has more freedom than the data it is fit to can constrain, and several "locks" are one-number calibrations of a curve whose shape is the actual content. The paper itself shows the symptom: six density laws give the same gradient (-0.00106 to -0.00109 Gpc^{-1}) because only ρ'/ρ at D is constrained, and the cluster data are matched equally well by laws with very different global behaviour.

Strengths

• **S1** The level-set lock is mathematically clean: one constraint fixes one scale per law; all six reproduce (M11–M13).

• **S2** The axisymmetric curve A_F is genuinely the CoE forward map, not a disguised fit (M16) — a fair test object.

• **S3** Good use of a $\Delta\chi^2$ profile (Fig. 9) to show the L–D degeneracy.

Weaknesses

• **W1 [MAJOR] Identifiability.** Location: §4.1, Figs. 8–9. The lock fixes only the local log-gradient; the cluster anisotropy fixes mainly a dipole + quadrupole phase. A 2-parameter family (D, L) is then being constrained by ~2–4 effective numbers. Fix: report the Fisher matrix / posterior for (D, L, shape, l_0) using a proper covariance and show which combinations are constrained.

• **W2 [MAJOR] Unspecified forward operator.** Location: Eq. 3 (f, p(s) unspecified, self-flagged). The core map from density to A cannot be reproduced from the paper (M9). Fix: give f and p(s) explicitly with code.

• **W3 [MAJOR] Free-form source in the CMB stage.** Location: Eq. 19, Table 2. A 12-parameter P_δ with two oscillatory terms and a free projection centre is a flexible basis; reaching $r^2=0.976$ shows the basis spans the TT shape, not that the model predicts it. Fix: a prediction test — fit $\ell<800$, predict 800–2500 (and TE/EE) blind.

• **W4 [MAJOR] Two sky orientations.** Location: §4.6 vs §4.7.2. Fitting uses +80.8° roll; anomaly comparison uses 15° lift + 45° rotation, unreconciled. Fix: one orientation, fixed before the comparison.

• **W5 [MINOR] D=42 vs 58.2** in the web modulation (M17) — the web is built for a different universe than the one the cluster fit selects.

• **W6 [MINOR] Pass/fail rule mislabelled** (R=8 PASS with lowl_over 2.99 > 2.5; R=14 FAIL unstated) (M25).

Alternative perspective

The same pipeline run with a pure ΛCDM + bulk-flow (or kinematic dipole + quadrupole) template on the cluster data would likely do at least as well (cf. M16: 3-harmonic $\chi^2=3.17$). The paper would be stronger as a demonstration that "an off-centre matter geometry is *not excluded* by cluster scaling data at the level of a low-order template", with the CMB stage separated out as an open problem.

Questions

1. How many effective degrees of freedom does the $A(\hat{n})$ curve have after the 90° cone smoothing?

2. Can the CoE model predict anything not used in calibration (e.g. the H_0 anisotropy amplitude in a different cluster sample)?

Devil's Advocate Report

Reviewer Information

• **Role:** Devil's Advocate (fixed seat). **Task:** find the strongest counter-arguments, logic-chain breaks, and alternative narratives. Neither pro-CoE nor pro-ΛCDM.

Overall Assessment

• **Recommendation:** Reject in current form (resubmit after the CRITICAL items are addressed). **Confidence:** 4/5. **Calibration:** NOT_CALIBRATED.

Critical findings

• **DA-1 [CRITICAL — data–conclusion mismatch] The cluster evidence does not discriminate.** The null model is not rejected even under the paper's own (over-optimistic) independent-bin statistic: $P(\chi^2 \geq 45.14|71)=0.993$. A free low-order harmonic fit beats every CoE law (3.17 vs 7.67–11.6), and the paper itself says the best fit equals "a reference sinusoid". With ~4 effective bins, "45→10" is not evidence for CoE over no anisotropy, nor over a bulk flow. (M15, M16)

• **DA-2 [CRITICAL — foundation collapse] CMB isotropy.** An observer 58 Gpc from the centre of a structure whose geometric CMB multipole is "several percent" violates the observed 1e-5 isotropy by ~100× even after the claimed 95.9 % cancellation (M43); this is the classic off-centre LTB failure mode. The cancellation figure itself is about the stochastic web, which the TT fit then re-tunes. Unless the full-sky low multipoles are shown to be ≲1e-5, the geometry is excluded.

• **DA-3 [CRITICAL — logic-chain break] Inconsistent backgrounds.** t(a) needs H≈47; the lock uses $H_0=70$; the web uses ΛCDM dark-matter parameters; R=4 Gpc needs $H_0\approx145$. The chain "geometry → lock → web → CMB" is not one model (M5, M6, M22).

• **DA-4 [MAJOR — stronger counter-narrative] Acoustic peaks.** The peak structure comes from cos(k r_s) terms inserted in Eq. 19 with r_s free; ΛCDM *predicts* peak positions, heights and damping from two densities. A flexible fit worse by $\Delta\chi^2\approx1{,}229$ (χ^2/dof 1.52 vs 1.03) is evidence against the model, not partial support.

• **DA-5 [MAJOR] Anomaly score is post hoc.** 11/12 after choosing orientation (different from the fitting orientation), smoothing, harmonic order and spot assignment, with antipodal pairs counted separately; no null simulation (M42).

• **DA-6 [MAJOR] The step that would test lensing/shear failed to reproduce** (run9: 48.9 vs 149.9) and is absent from the paper (M26, M44).

Strongest counter-narrative

The Migkas anisotropy is a low-order (dipole/quadrupole) pattern of modest significance that Migkas et al. themselves attribute possibly to a bulk flow or systematics; CatWISE's excess dipole is under active debate for kinematic/systematic reasons. Both can be accommodated by templates with fewer assumptions than an off-centre 58-Gpc matter structure, and the CMB, which is the most precise isotropy test, is fit better by ΛCDM with fewer parameters. Nothing in the manuscript discriminates CoE from these.

What would change my view

(1) A full-sky CMB low-ℓ prediction ≲1e-5 from the final model; (2) a cluster likelihood on the 313-cluster catalogue with Δ(BIC) favouring CoE over a bulk-flow template; (3) a blind out-of-sample prediction (high-ℓ, TE/EE, SN D_L, or a new cluster sample) that succeeds.

Fairness check

Things the paper gets right: arithmetic (nearly all checks in Appendix M verify), honest self-flags, honest statement of the CMB shortfall. The criticism is of inference and physical consistency, not of the author's honesty.

Phase 0 — Field Analysis & Reviewer Configuration (field_analyst_agent)

Paper Basic Information

• **Title:** Off-centre CoE cosmology: the mathematical workflow from geometry to the Planck TT spectrum (rev. 5, compiled 24 Oct 2026)

• **Source:** /workspace/coepdf/main_rev5.tex (431 lines), PDF /workspace/coe_math_workflow_rev5.pdf

• **Type:** methods/working-notes manuscript; phenomenological model + fits to three data sets (Migkas L_X–T anisotropy, CatWISE dipole, Planck 2018 TT).

• **Maturity:** working draft. Self-flags ([unverified: ...]) present; references incomplete (no years/journals for most entries); two sky orientations unreconciled.

Field Analysis

Dimension	Assessment
Primary discipline	Physical cosmology (inhomogeneous / off-centre cosmologies, CMB phenomenology)
Secondary	Observational X-ray cluster cosmology; large-scale-structure statistics; inverse problems / statistical model checking
Paradigm	Quantitative, model-fitting, exploratory
Methodology type	Phenomenological forward modelling (Newtonian + FRW thermal history), χ^2 minimisation, harmonic residual analysis, Monte-Carlo random webs
Target tier	As written: arXiv preprint / working notes. With the issues below fixed: MNRAS, PRD, JCAP (all routinely publish off-centre LTB and anisotropy tests)

Recommended Target Journals (Top 3)

1. Phys. Rev. D — standard home for LTB/off-centre-observer CMB constraints (Alnes & Amarzguioui 2006; Foreman et al. 2010).

2. JCAP — anisotropy/dipole-tension papers.

3. MNRAS — Migkas-type cluster anisotropy follow-ups.

Reviewer Configuration Cards

Card #1 — Journal-Fit Reviewer (EIC)

Editor of a PRD/JCAP-type journal handling non-standard cosmologies. Focus: is the claim novel against the LTB-void literature, is it framed honestly, is it a paper or notes.

Card #2 — Reviewer 1 (Methodology)

Cosmological statistician (likelihoods, correlated binned data, model comparison, reproducibility). Focus: χ^2 on overlapping sliding cones, dof accounting, look-elsewhere in the 11/12 anomaly test, ensemble-mean vs realisation, code-to-number traceability.

Card #3 — Reviewer 2 (Domain)

CMB / inhomogeneous-cosmology theorist (LTB, Sachs–Wolfe, lensing, kinetic SZ). Focus: CMB isotropy bounds on off-centre observers, GR validity at $D \approx 58$ Gpc, consistency of using ΛCDM inputs (Ω_m incl. dark matter, Eisenstein–Hu CDM transfer) inside a "no dark matter" model, literature coverage.

Card #4 — Reviewer 3 (Perspective)

Applied mathematician in inverse problems / geophysical tomography. Focus: identifiability of m(r), degeneracy, regularisation, whether residual-harmonic peeling is diagnostic or just flexible curve fitting.

Seat #5 — Devil's Advocate (fixed)

No card. Attacks core thesis and logic chain.

Review Strategy Recommendations

• Methodology and DA seats carry the most weight (statistics on Migkas bins and the CMB stage are where the claims rest).

• A separate mathematics check (re-derivation + numerical spot check against box code/data) is attached as Appendix M.

• Provenance note: dispatch of independent subagents was unavailable; the five Phase-1 reports were written sequentially by one executor, each written to file before the next began. Role separation is therefore procedural, not independent error processes (as the skill itself states).

• Calibration status: NOT_CALIBRATED.